

Welcome to Code Crunch!

C5 | CRUNCH LABS: AI & Data Science

[Mon 1-2PM, Wed 12:30-1:30PM]
Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



CODECRUNCH.CS.FIU



Unit #6

Association Analysis

C5|Crunch Labs: Crunch AI Data Science

Unsupervised Learning

What is Unsupervised Learning?

- model is trained on data that has no labeled outputs.
 - i.e., the data used to train the model does not include predefined categories or target values.

When is an instance Unlabeled?

- an instance is unlabeled if there is no output feature value for that instance

why?

- The main objective is to explore and model the underlying structure of the data
 - such as grouping similar data points or identifying patterns.

Records

Age	Gender	Income	Profession	Tenure	City
35	M	60,000	IT	12	KRK
23	F	90,000	Sales	3	WAW
18	M	12,000	Student	1	KRK
42	F	128,000	Doctor	13	KRK
34	M	63,000	Manager	8	WAW
56	M	82,000	Teacher	30	WAW

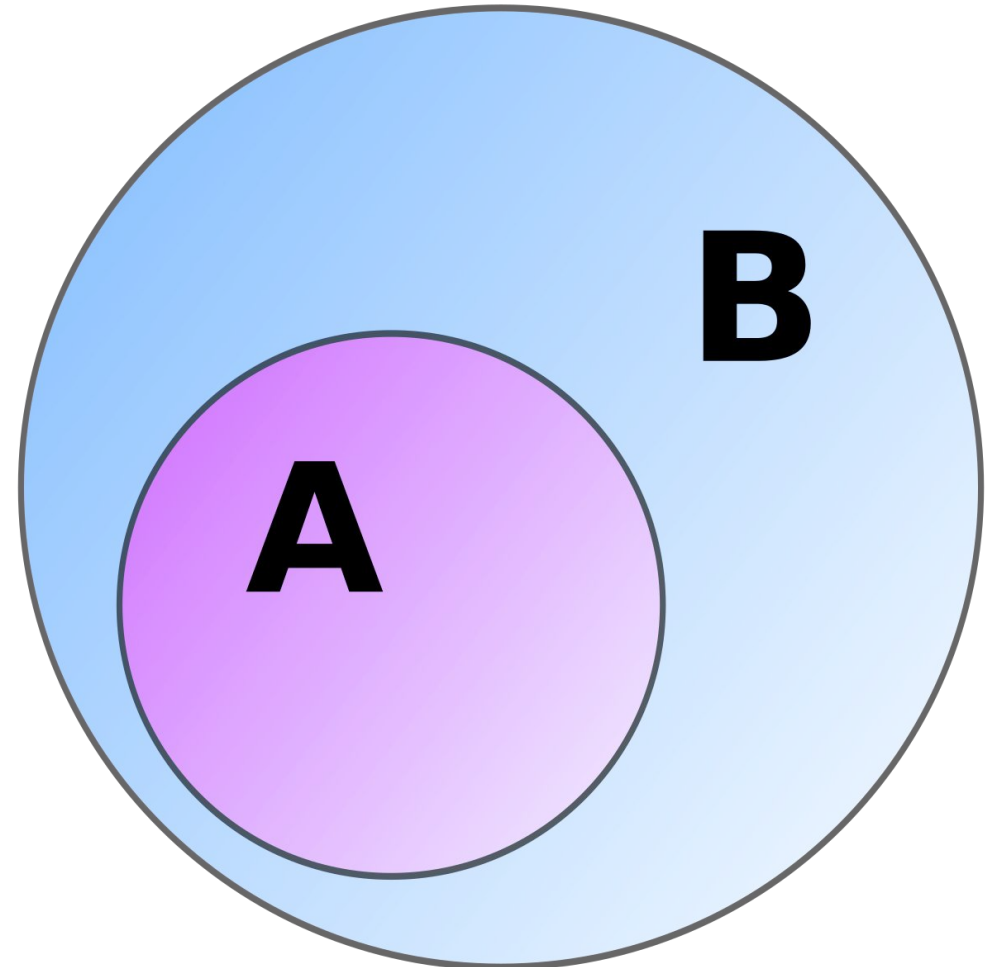
pre-knowledge

Sets

- A **set** is a fundamental data structure in mathematics and computer science that represents a collection of **unique** elements.
 - Unlike lists or arrays, sets do not allow duplicate values and typically do not maintain a specific order.

Subsets

- A **subset** is a set that contains some or all elements of another set. If every element of set **A** is also in set **B**, then **A is a subset of B**



Association Analysis

What is Association Analysis?

- Association analysis is a data mining technique used to identify relationships, patterns, or correlations between items in large datasets.
 - It is widely used in market basket analysis, where businesses analyze customer transactions to discover which products are frequently bought together.
- **key concepts**
- **Support:** Measures how frequently an itemset appears in the dataset.
- **Confidence:** Measures the likelihood of one item appearing given another item.

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}
11	{A}

CONT.



Association Analysis Algorithms

- Apriori Algorithm
- FP-Growth

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
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11	{A}



Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch